Catastrophic bifurcation transitions on financial markets

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Abstract

We provide empirical and phenomenological evidence for the occurence of catastrophic bifurcation transitions on notoriously unpredictable financial markets [1]. In particular, we observe and characterise the flickering phenomenon, which is a marker of the *catastrophic slowing down* analogous to the phenomenon of the critical slowing down in physical systems. We develop metrics for characterising the dynamics of the *catastrophic bifurcation transition* and evaluate their applicability in detecting an approaching bifurcation transition on financial markets of small and middle to large capitalisation.

Keyword: Dynamical phase transition, Catastrophic bifurcation transition, Catastrophic slowing down, Critical slowing down, Financial market

1 Catastrophic slowing down

Discontinuous phase transitions and critical phenomena in complex systems such as in liquid-gas systems are of canonical importance in statistical thermodynamics and its applications. *Critical slowing down* phenomena, observed in physical systems approaching critical point of the continous phase transition is often accompanied by instabilities and breakdowns. Sufficiently far from the critical point, metastable phases may form and vanish in minuscule discontinous phase transitions. Such microscopic breakdowns lead to macroscopic 'flickering' which is usually observed on approach to criticality. A host of other macroscopic phenomena accompanies and signals approach to criticality in a dynamical setting where the system may evolve arbitrarily close towards a globaly critical state.

Such early-warning signals have recently been identified and elucidated to be a consequence of a catastrophic bifurcation transition phenomenon observed in multiple physical systems, e.g. in the US housing market, medicine (epileptic seizure, asthma attack, depression onset and termination), and in climate dynamics [2, 3, 4, 5, 6, 7]. While conceptually appealing, in some cases evidence remains disputable [8].

Here we propose metrics sensing a catastrophic bifurcation transition on financial markets. Using these mentrics we examine financial markets of small (e.g. WIG) and middle (e.g. DAX) to large capitalization (e.g. DJIA) using daily return data.

Empirical data shown in Fig. 1 give: (i) $-1 < \lambda \leq 0$ and (ii) value of $-\lambda$ tends to zero (from its positive side) vs. time (counted in trading days (td)), forming a local absolute minimum. This vanishing minimum is a necessary requirement and a signature of the catastrophic slowing down phenomena and the catastrophic bifurcation transition's existence. The quantity $-\lambda$ can be interpreted as the temporary recovery rate.

2 Flickering precursor & catastrophic bifurcation

Figure 2 empirically shows that bistability can exist on a financial market, as a consequence of the fold topology of the catastrophic bifurcation transisition. The so-called flickering phenomenon can be observed before the negative catastrophic spike (cf. Fig. 2). Such unstable intermediate states

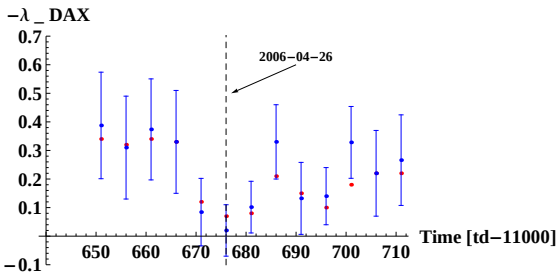


Figure1: The temporary recovery rate $-\lambda(\geq 0)$ calculated for DAX by using two different formulas: (i) $-\lambda = 1-AR(1)$ (full blue circles with bars) and (ii) $-\lambda = 1-ACF(1)$ (full red circles). The Formula (i) gives the relation between the recovery rate, $-\lambda$, and the linear autoregression coefficient, $AR(1)$. The Formula (ii) gives an analogous relation between autocorrelation, $ACF(1)$, of the subsequent daily values of DAX time series. Both curves are concave, having broaded minima self-consistently located at the same time denoted by the vertical dashed line. This line denotes the position of the negative spike (shown in Fig. 2), which defines the onset of the catastrophic bifurcation transition.

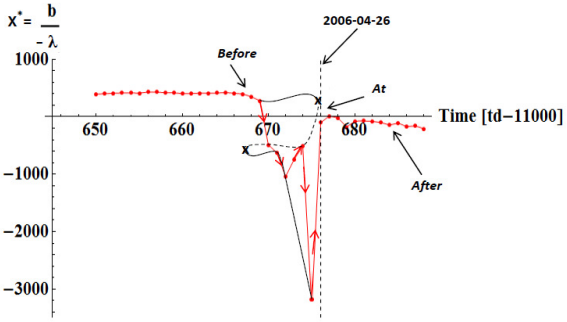


Figure2: DAX empirical fold bifurcation curve depicted with red points joined by red segments of lines and arrows. It captures a sequence of "mechanical" equilibrium states parametrized by $x^* \stackrel{\text{def.}}{=} -b/\lambda$) vs. time, with b and λ being the only parameters of the linear autoegression; b is the temporal shift parameter. The dashed line between the two tipping points (denoted by crosses) identifies a sequence of unstable "mechanical" equilibrium states. Both the 'Before' and 'After' sequences consist of stable "mechanical" equilibrium states. The 'At' arrow indicates a possible catastrophic bifurcation threshold. Prior to this threshold (indicated by the dashed vertical line at 2006-04-26) flickering is clearly observed.

mainly appear on approach towards the bifurcation regime. As such, the flickering phenomenon can be considered as a possible precursor of an imminent catastrophic bifurcation transisition.

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